Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

WEEKLY LESSON PLAN

Department: Computer Science and Engineering

Year / Semester: 2025-26 / Winter

Course Code / Course Name: 10213CS224 / Full Stack Application Development

Unit III: Advanced Spring, Microservices Programming **Course Outcome: • Utilize micro services architecture for building scalable applications**				
Week	**Session**	**Topics to be Covered**	**Problems to Practice (Hands-On)**	**Resources**
Week 7	Session 31	Introduction to Microservices, Single Responsibility & High Cohesion	**Task 7.1:** Design microservices for a **Library Management System** based on the Single Responsibility Principle. List at least **four services** and one responsibility for each.	https://www.geeksforgeeks.org/system-design/microservices/
	Session 32	Loose Coupling & API Communication	**Task 7.2**: Design a simple interaction between Order Service and Payment Service using API communication. Write the steps involved in the interaction.	https://www.youtube.com/watch?v=nfkdKHcKxbE
	Session 33	Independent Deployment & Decentralized Data	**Task 7.3:** Design microservices for a **Hospital Management System** by following decentralized data management. Mention at least **three services** and their respective databases	https://www.youtube.com/watch?v=ewUw0sUxHI4

	Session 34	Service Registry and Discovery	**Task 7.4**:. Online Shopping Application: • User Service • Product Service • Payment Service Each service registers with the service registry. When the User Service needs Product Service, it discovers it through the registry instead of using a fixed address.	https://www.youtube.com/watch?v=wC2eTPoQ1GI
	Session 35	Develop a micro service-based system	**Task 6.5**. **(syllabus task 14):** Design a simple microservices-based system by splitting a monolithic application into at least two independent services. Apply microservices design principles such as loose coupling, single responsibility, and independent deployment.	https://www.youtube.com/watch?v=rv4LlmLmVWk&t=198s